Question	Answer
Can you explain how mother can be LOSS?	The mother can explain to the kid whether they did good or not. For example, did they do good or bad in putting the sqaure piece into the correct position. The backpropagation analogy will be telling the kid "why and how" they were wrong (loss), in case they did take a wrong step.
Is there a mechanism where the ML verifies Unsupervised learning theorem with supervised learning? Is the such a concept?	Unsupervised learning is used where the output label is not present, so the supervised learning cannot verify due to absence of labels. However, we can remove the labels for the unsupervised learning and use the labels to verify using supervised learning, if needed.
Can I assume AI is typically reinforcement learning at a larger scale?	AI is not always reinforcement learning. If the use case/problem that we are trying to solve reinforcement learning, it's chosenonly then. If the problem can be solved with models with lesser complexity, we opt for that
How can you differenciate unsupervised learning adn reinforement learning? It seems both are unsupervised	Unsupervised learning passively analyzes data to find patterns. RL actively interacts with an environment to maximize a reward.
what is difference between unsupervised learning and reinforcement learning	Supervised learning is when there is a loss defined, and there is also a mechanism to tell "how and why the mistake happened". Unsupervised learning is when an external loss is not defined, so the model or algorithm itself comes up with a loss (according to our needs) and learns "how and why the mistake happened". In reinforcement learning, there is only the external loss, without any information of "how and why the mistake happened", so the learning happens only through taking the decisions with positive and negative rewards for a long time. As there is no mechanism of backpropagation (that is, information of exactly what happened wrong), reinforcement learning requires much more time and iterations compared to supervised and unsupervised learning.
You give labelled data sets – based on which it identifies patterns and build the AI model what happens one layer below – how is the AI model build ?	AI models are also built in a similar way. The models have been trained on very large datasets, often on datasets that are publicly available(like wikipedia or other open internet sources).
What is the corresponding components in a computer to brain and neurons in humans?	In an AI/ML context, we usually map this out conceptually (in software) rather than physically: The Brain corresponds to an Artificial Neural Network (ANN). A Biological Neuron corresponds to an Artificial Node (or Perceptron). Synapses (connections) correspond to Weights (the mathematical values determining connection strength).

In artificial neural network - can you explain what is neural here, in an example?	The word “neural” in the phrase artificial neural network refers to the fact that they are inspired by biological neurons. An artificial neuron consists of inputs, weights, a bias, and an activation function.
What is a practical real-life example that defines the difference between unsupervised learning and reinforced learning?	Let's use a smart robot vacuum to see exactly how both operate differently in the same environment. Unsupervised Learning (Finding Patterns): The vacuum's camera scans the floor and naturally groups the debris it sweeps up into distinct clusters based on size and shape (e.g., fine dust, pet hair, and dropped cereal). It doesn't have labels for what these items are; it just mathematically recognizes, "these items are similar to each other." Reinforcement Learning (Taking Action): The vacuum learns “how” to physically navigate your living room. Though it has a map of your room, the map doesn’t carry any details about non permanent objects(a random chair, a pet dog etc.). It uses RL to learn the positions of objects through reward and penalty mechanism
is neural means the feature?	neural in an artificial neural network (ANN) means that the ANN is inspired from neurons (the brain cells) in operation.
How is the Neural Network related to Supervised or Unsupervised learning?	Neural Network is a model architecture while supervised and unsupervised learning describe how the model learns from data
ok, you mean -> y = mx + c here neural means combination of m & c?	Yes, basically m and c can be learned by the neuron to output correct y value when x value is given as input.
is synapse correspond to memory in ANN? and it is for short time right?	ANN are artificial neural networks. The weights in ANN can be thought of as analogous to biological neurons, storing parts of memory. The concept of short time and long time does not come here, and is an advanced topic (large language models) that will be covered in later lectures. For now, just know that the model has weights, and takes inputs and produces outputs, effectively acting as a system. For example, in a system of y = mx + c, there is no concept of short term or long term memory.
What is the role of pathway in ANN example mentioned?	Think of the pathway as a road through the network. The neurons are the processing points and the weights determine how strongly information travels along each road.
what does 10 power -1 mean with respect to number of neurons	10 power −1 means 0.1. It does not indicate the number of neurons. The number of neurons is specified separately as an architectural parameter.

Like Linearr and Logistic Regression, is ANN and CNN not a separate model themselves?	Yes, ANN and CNN work differently and are separate models. More on this in later lectures.
sponge has 10 to the power -2 what does this mean with repect to number of neurons	Sponge has 0 neurons. Here it is put as 10^(-2) = 0.01, meaning it basically has no neurons. More accurately, sponge actually has 10^(-inf) = 0 neurons, which is difficult to show on a log graph.
How the threshold got calculated?	The threshold for each class is typically selected on the basis of the model output and the desired goal of the classification. For example, the threshold for binary classification is 0.5 by default but it might be increased or decreased in order to increase the precision, recall, and other metrics based on the problem at hand
if kid is learning threashold, how w1, w2 is decided? are we born with it?	w1 and w2 are learned by making mistakes for a long time. The more mistakes the kid makes, the more they understand what w1 and w2 should be and should not be.
why w1,w2 and threshold require	Imagine that you are buying a house: Think of a single neuron making a decision, like: "Should I buy this house?" Inputs: The raw facts. (like “Is it near my office?”, "Is it within budget?”). Weights: How much you care about each fact. If a short commute is a total dealbreaker for you, w1 gets a massive value. If your budget is somewhat flexible, w2 gets a smaller value. Weights act as a volume knob, amplifying or muting specific inputs.Threshold (Bias): Your baseline hesitation to buy a house at all. Even if the house has a great commute and is cheap, your total "pro-buying" score needs to be high enough to overcome the massive financial commitment (the threshold). If the weighted inputs cross this threshold line, the neuron gets triggered (you decide to buy).
How threshold is decided?	The threshold for each class is typically selected on the basis of the model output and the desired goal of the classification. For example, the threshold for binary classification is 0.5 by default but it might be increased or decreased in order to increase the precision, recall, and other metrics based on the problem at hand.
What is Neurode?	A "neurode" is an early term used to describe an artificial neuron, prominently featured in the 1943 McCulloch-Pitts model. It acts as the central computational unit that takes in incoming signals or inputs (like x1 and x2), applies their respective assigned weights (w1 and w2), and produces a final output based on whether a specific mathematical threshold is met. In modern machine learning terminology, this exact same concept is typically just referred to as an "artificial neuron."
Sir What is W2 here in Mcculloh pitt equation?	W_2 is the weight associated with the second input X_2. It determines how strongly X_2 contributes to the neuron's output.
what does MX and represents here/	MX represents the weighted input i.e. weight x input

what is the formula calculated for this threshold.	There isn't a fixed mathematical formula to calculate the threshold (often called the bias, or the ‘c’ in your y = mx + c equation) upfront. Instead, the neural network learns it automatically during the training process. For exacmple, think about finding the perfect temperature in a new shower. You don't use a formula to calculate the exact angle to turn the knob. You start with a guess, feel if it is too hot or cold (checking the error), and nudge the knob slightly (adjusting the threshold) until it feels just right.
If W1 and W2 are weights then how exactly we are calculating the weights?	we don’t calculate the exact value of weights like w1, w2 before hand. It’s more of a repetitive process of trial and error wherewe trymultiple values for weights and eventually arrive at the correct value. Like you decide the optimal value ofyour shower temperature by adjusting the knob
who defines the threshold?	When designing the function, the user can set the value. If the neuron learns, then the threshold is actually learned by the neuron.
can any threshhold be 0 ?	Yes, a threshold can be 0. It depends on how the model is defined. There is no requirement that the threshold must always be positive.
How does the threshold is always zero in this case, i missed completely ?	There is a w_0, which is basically the threshold. So instead of saying for example y = 1 when mx >= b and y = 0 when mx < b, we are saying y = 1 when mx - b >= 0, and y = 0 when mx - b < 0. w_0 = -b is called the "bias".
what is bias here ?	Let’s say you are considering to buy a house. What is the Bias (Threshold)? The bias is your baseline hesitation to buy any house. It is a constant value added to the equation to shift the baseline requirement up or down. Even if the house has a great commute and is cheap, your total "pro-buying" score (the inputs multiplied by their weights) needs to be high enough to overcome the massive financial commitment—that built-in hesitation is the bias.
what is a bias?	Bias is like an intercept in a linear equation. It allows the model to adjust the output independently of the input values
and in Y = mx+c , what does c represent and Y	c is the bias and Y is the predicted value.
how bias and threshold are different. first sir told bias is nothing but threshold and second statement sir said through bias capturing threshold	Bias here basically shifts the threshold to 0. Threshold is a value that defines a decision boundary, while bias shifts the function/system, effectively shifting the threshold to 0.
threashold & bias both same?	No, they are not exactly the same. Threshold decides when the neuron activates while bias shifts the decision boundary.

Why this is “+w0” rather “-w0”	It is purely a mathematical convention to keep the equations clean and uniform. The network does not actually care whether you program it as +w0 or -w0. If you write +w0 but the neuron inherently needs a negative threshold to function correctly, the optimization algorithm will simply learn a negative number for w0 (e.g., w0 = -5). Example: It is like programming a generic "adjustment fee" on an invoice as Total + Adjustment. If you need to give a discount instead of charging a fee, the system just learns to set the adjustment to -20. You do not need to hardcode a separate minus sign into the software.
threshold or bias is same as error term??	No. Threshold or bias is a parameter of the system, not to be confused with any type of error. For example, if the function is y = mx + c, c is the "intercept", and the function changes if c is mistakenly treated as error term and c is removed.
It was - bias earlier why is it + bias now	It can be either positive or negative, and it absolutely does not matter when you start. The network will automatically adjust it to whatever sign it needs during training.
Hi. If we can slow down and take W1X1 + W2X2 through a real life example to create a permanent image in mind, would have been really good to then grasp the incremental concepts. Thank you for understanding	Lets sayyou are trying to predict house prices right you have two features (Area)X1 and (crime rate in your locality)X2, now if we train this W1 would be some high positive value as area increases price increases, and W2 would be some negative value as more crime rate means less price
in last class, one of the "Take home" problems asked which of the 3 problems can be solved using linear classifier. Option A can be solved, but not Option B and C. can anyone confirm?	True, and option B is somewhat similar to XOR problem. Glad you tried them :)
What is meant by bias	Bias is an additional parameter added to the weighted sum. It acts like the intercept and helps the model shift the decision boundary.
the value of w1 and w2 cannot be zero?	No, w1 and w2 can be zero. A zero weight simply means that the corresponding feature has no contribution to the prediction.
How does a Function derived in case of Linear Model with reference to independent and dependent variable, what kind of logic is used?	In linear regression, we assume a linear relationship: Y = wX + b. The model learns w and b by minimizing the difference between predicted and actual Y usually using the least-squares principle.
I think we might not need bias if we normalize the dependant i.e. Y crossing origin.but if we want to keep Y same, we will add bias	Yes, your understanding is correct. If the data is constrained to pass through the origin, the bias can be zero. Otherwise, the bias allows the model to shift the line vertically and fit the data better.
Is bias same as intercept?	Yes. In linear regression, the bias is equivalent to the intercept c in Y=mx+c. Both represent the predicted value when all input features are zero.
is weight same as parameters	Yes. In a neural network, weights are parameters learned during training.

From where the weight comes during modeling.	Weights are initially assigned randomly and then learned during training using backpropagation and gradient descent.
every nueron is a virtual machine in actual AI system?	No. A neuron is a mathematical computation not a virtual machine. Many neurons are implemented through software operations on CPUs/GPUs.
The 'c' in integration, can also be called as bias?	Conceptually, they are remarkably similar! Both act as an anchor that shifts a mathematical function up or down without changing its fundamental shape. However, the reason they exist is completely different: In Integration (The Unknown): When you take a derivative, constants turn into zero (information is lost). When you integrate backwards, you add '$+ C$' simply to say: "There might have been a starting baseline value here, but I don't know what it was." In Machine Learning (The Tool): The bias is not a missing piece of information; it is a deliberate, learnable tool. We add bias so the model has the flexibility to shift its line or curve away from the origin (0,0) to wherever the actual data lives. The takeaway: Visually, they do the exact same thing (shift a curve vertically). But philosophically, integration's 'C' is an unknown placeholder, while ML's bias is a parameter the algorithm actively learns and tunes.
how the value of bias decided? is it always >0?	Bias is learned during training. it can be positive, negative or zero. It is not always > 0.
With y' = b + w1x1, if bias keeps changing, will it be dependent on x(input)?	No, the bias is never dependent on the input ‘x’. It is a completely independent baseline. If the bias is changing, it is only because the model is still in the training phase and is trying to find the best baseline. Once the training is complete, the bias locks in as a single, fixed number for all future predictions.

what is the need of using bias why always one bias in an equation? how does bias and threshold differ?	Bias shifts the decision boundary. Without bias (y = mx), your prediction line is mathematically forced to cross exactly through the origin (0,0). Bias (+ c) gives the line the freedom to shift up or down to actually reach where your data lives. Why only one bias per equation? Because basic algebra combines them. If you tried to add a +2 bias for age and a +3 bias for salary, the equation just collapses them into a single +5 shift. One neuron draws one line (or plane), so it only needs one overall baseline shift. How do bias and threshold differ? They are the exact same concept
Is Leaner model use forecast the next month sales?	Yes. A linear model can be used for forecasting future sales if the data has a suitable relationship with the predictors.for time-series data, dedicated forecasting models are often used.
what can be limit of bias?	Bias has no fixed limit. it can be any real value - positive, negative, or zero.
Is slope and weights same in linear regression equation?	Yes. In a linear equation, the weight represents the slope i.e. the change in output for a unit change in input.
did not understand the XOR problem, like what those lines represent in plane	They represent decision boundary, basically to separate 2 classes you need to draw a line in case of 2D data, now because in case of XOR((1, A), (2, B), (3, A)(4, B) where A and B are two classes. So no linear line could separate them
Does hyperplane separates only two classes? What if there are multiple classes?	A hyperplane can separate two classes directly. For multiple classes, we use multiple hyperplanes such as in one-vs-rest or softmax classification.
what is the need for bias , what if we dont have Y ?	Bias lets the model shift the output independently of the input. If there is no Y, there is nothing to predict.Bias is still a model parameter but is learned only when training has a target.
what is the need for bias , what if we dont have intercept ?	If there is no intercept, we actually do not need bias. However, using bias always is better, because in general the intercepts are expected to be there. If no intercept is there, the bias term can just converge to 0 value.
hoq do we decide on bias at initial or start before it learns, is there a way we change bias as learning progressing	Bias is also learnt along with weights. Weights and biases together are called "learnable parameters".
how to decide upon the linear/polynomial model to use for regression?	Depends on the dataset, if the dataset is linearly separable or not
How will the input be non-linear be represented	If you are asking about the case when the output is non-linearly dependent on the input, a linear model in fact cannot fit properly to the problem, and will not perform good. In that case, we can use polynomial regression.

How can I see these mathematical values(gradient, slope, bias etc) while training my model?	The gradients/slope, weights and biases, all change as the learning progresses. So the values can be stored and plotted accordingly after each learning iteration.
what is the need for bias , what if we dont have intercept ?	Bias acts like the intercept. Even if we don't explicitly have an intercept, the bias provides that shift in the equation.
what is meant by fitting a non linear to linear y = ax2+bx+c so we say x' = ax2+bx then y = dx'+c ?	Means that we transform the features x1, x2 changes to x1, x2, x1^2, x2^2, x1.x2, the classifier is linear, but because we are modeling non linear relationships, the decision boundary is non linear
what is the need of using bias why always one bias in an equation? how does bias and threshold differ?	this of it this way without bias equation of hyperplane/decision boundary is w1x1+w2x2 correct?Now this will always pass through origin, That's why we add b and equation becomes w1x1+w2x2+b now w1, w2, and b could be learned
I'm not expert in Mathematics and learnt math quite long back. So is all this mandatory in ML?	No, you don't need advanced math. Basic algebra, graphs, probability, and statistics are enough to understand most ML concepts.
why there is only overfit and no underfit in Von -Fermi equation?	The von Neumann–Fermi equation typically focuses on the model fit side.underfitting is a separate concept that occurs when the model is too simple to capture the data pattern.
I didn't get what loss is. Could you please explain with simple terminology	The loss is an information of "how bad the model is", that tells the model that the model needs to improve. For example, say a model sees one cat and one dog. It classifies both as cat. Then, it got 1 right and 1 wrong. So this information of "1 wrong" can be informed to the model so that it can learn further to improve its performance.
@Muazul Hasan Khan, got it, so we create multiple dimension? but how can we sure that x1, x2 are othogonal? for ex: x, x2, are they orthogonal?	No thats not important. Think of it this way even if two features x1 and x2 are highly correlated, we could still draw a linear decision boundary(though its not a best practice)
When we use the term "loss", is it for every Q&A for an AI model? or based on an area e.g. social science Q&A? What loss actually means for a layman to understand?	"Loss" means how far the model's prediction is from the actual answer. The exact loss function depends on the problem, e.g. MSE for regression or cross-entropy for classification.
Why we use MSE ?	MSE is used because it penalizes larger errors more and is easy to optimize mathematically.

What is MSE(THETA) here? and what will be value of that?	In Ridge Regression, the goal: Minimize (MSE + Penalty Term). The Penalty Term (L2 Regularization):Ridge regression takes your standard MSE and adds a "tax" based on the size of your weights (specifically, the sum of the squared weights). Why do this? If a model has massive weights, it becomes too sensitive and overfits the training data (memorizing noise). By adding this penalty, Ridge regression forces the model to keep the weights ('theta') as small as possible while still trying to fit the data.
what is MSE	MSE is mean squared error. Say the actual output should be 1 and 2. The model predicts 0 and 4 corresponding to 1 and 2 desired outputs. Then, MSE is ((1-0)^2 + (2 - 4)^2) / 2 = (1 + 4) / 2 = 2.5. The MSE loss decreases as the model becomes more accurate.
I have a dataset with less than 0.7 % of labelled data for supervised learning. The proposed features are not effectve to demarcate between the labelled and non labelled rows in the training data. Now the model training is dominated by categorical attributes and the result kpi's are not encouraging. What steps can be done to imporve this istuation?	So to begin with you train a model with 70% dataset that you have find prediction on the 30% unlabelled dataset, and add the samples to the train set for which you initial model was very confident this is semi supervised learning technique
Can you explain the cost function?	cost function measures the overall error of the model. Training tries to minimize this cost
In lasso regression MSE is loss or error?	In Lasso regression, the Mean Squared Error (MSE) acts as both a measure of error and the primary component of the loss function. As an Error: MSE calculates the average squared difference between the actual target values and the values predicted by the model. It tells you how far off your predictions are. As a Loss: In the optimization objective of Lasso regression, MSE serves as the loss component (representing model fit), which is added together with an L_1 penalty (absolute values of the coefficients) to form the complete loss function that the algorithm minimizes
Also suggest the ways to handle huge training data in absence of enough cores and memory. What options or methods do we have?	Use mini-batch training, data generators and distributed processing. Also use efficient data types and load only the required features instead of the entire dataset into memory.

If an company say Claude has to comment on the loss function of Opus 5.5, how will they calculate it? Or for any company for their models?	This is outside the scope of this module. But to answer your question LLMs model generation as a classification problem, they generate tokens, at each time step the probabilities corresponding to all the tokens in the vocabulary is generated, but from the ground truth theres only 1 token at each timestep. So we calculate cross entropy loss. This is the scenario for training. Later evaluation of LLMs are task specific.
L2 regularizer should be used for data w outlier since it squares the losses . is the assumption correct ?	Squaring actually makes a model highly sensitive to outliers, not robust to them. If a normal data point is off by 2, the squared error is 4. If an extreme outlier is off by 10, the squared error explodes to 100. This massive penalty forces the model to aggressively bend the line toward the outlier just to reduce that giant number, which ruins the fit for all the normal data.
Sir, why does L1 regularization make some coefficients exactly zero, while L2 usually only makes them small but not exactly zero?	It comes down to the formula used. When the parameters are large, squaring the term makes it very large, so L2 penalizes heavily compared to L1. When the parameters are small, squaring the term makes it very small, so L2 penalizes very less compared to L1. L1 does not differentiate between larger and smaller parameters, while L2 penalizes larger parameters more and is relaxed for very small parameters.
What does w mean here? the prarams	Yes. w means the model weight assigned to a feature. During training, the algorithm learns the best values of these weights.
can we say L2 is tending to 0, and L1 is likely to be 0?	Yes, with a small correction. L2 tends to make weights smaller and closer to zero, while L1 can make some weights exactly zero. It doesn't mean all weights become zero.
One more queston - is L1 si similar to MAE?	They both use the absolute value but for different purposes: MAE measures prediction error L1 penalizes model weights So mathematically similar but conceptually different.
while explaining the formulies/functions can you try to connect those with AI in general language. How are those being used in AI?	As a real-life example, let’s look at self-driving cars. When an AI predicts the distance to a pedestrian, it relies heavily on loss functions like MSE. If the car predicts a person is 10 feet away but they are actually 2 feet away, squaring that 8-foot error turns it into a massive numerical alarm (64). This forces the car's neural network to aggressively correct its perception weights so it never makes that dangerous mistake again.
When do we use Ridge and when we use Lasso?	We use ridge when we want to keep all features but reduce their imact especially when features are correlated. Lasso is used when we want feature selection because it can make some coefficients exactly zero

Lasso means do the task then adjust to zero	Lasso adjusts the weights to zero during the task, not after it.
if I want the stability of Ridge as well as the feature selection property of Lasso can we combine L1 and L2 regularization?	Yes and that’s known as elastic net regression.
Is there any guidelines to decide between L1 & L2?	Use L1 (Lasso) for Feature Selection: Choose this when you have hundreds of inputs and suspect many are useless noise. L1 aggressively pushes the weights of irrelevant features to exactly zero, effectively deleting them and leaving you with a simpler, cleaner model. Use L2 (Ridge) for Feature Shrinkage: Choose this when you believe most of your inputs are actually useful, but they might be highly correlated (e.g., predicting house prices using both "square footage" and "number of bedrooms"). L2 shrinks all weights evenly to prevent overfitting, but keeps every feature in the final equation.
Why does L1 norm always needs weights to be 0? what is the significance?	For that we will need to look at the visualisation of loss contours but to explain it verbally L2 pushes harder when a weight is big and gently when it's small, so as a weight gets close to zero, the push almost disappears. The weight gets tiny but never quite hits zero. L1 pushes with the same force no matter how small the weight is. If a feature isn't useful enough to fight back against that constant push, its weight gets pushed all the way to zero.
Can ridge be a best option for forecasting related use cases compared to Lasso?	Yes. Ridge is often useful when forecasting has many correlated features such as lag variables. It keeps those features but shrinks their coefficients. Lasso may remove some of them entirely.
Any real life example where to use ridge and where to use lasso?	-> Use Lasso (L1) when you want automatic feature selection because it forces less important feature weights to exactly 0. Real-life example: Genetic research, where you have thousands of gene expressions but want to isolate the few specific genes that actually predict a disease. -> Use Ridge (L2) when you want to keep all your features but shrink their weights to prevent overfitting, especially when features are highly correlated. Real-life example: Predicting house prices, where features like square footage, number of bedrooms, and number of bathrooms are heavily correlated but all matter for the final price.

Can you give a practical real-life example when to select Ridge and Lasso? What it means: Lasso is happy with zero weights. Ridge is happy with less weights?	-> Use Lasso (L1) when you want automatic feature selection because it forces less important feature weights to exactly 0. Real-life example: Genetic research, where you have thousands of gene expressions but want to isolate the few specific genes that actually predict a disease. -> Use Ridge (L2) when you want to keep all your features but shrink their weights to prevent overfitting, especially when features are highly correlated. Real-life example: Predicting house prices, where features like square footage, number of bedrooms, and number of bathrooms are heavily correlated but all matter for the final price.
got it. So for outlier we should rely on L1 norm	Yes, but only if you mean L1 Loss, not L1 Regularization
can you give quick example of overfitting and underfitting? Also, why did we focused only on overfitting in current example?	-> Overfitting is like a student who memorizes the exact questions on a practice test but fails the final exam because they cannot generalize to new questions. In machine learning, this happens when a model is too complex and memorizes the training data, resulting in high training accuracy but much lower test accuracy. -> Underfitting is like a student who barely studies and fails both the practice test and the final exam. The model is too simple to capture the underlying patterns, leading to poor performance on both the training and testing sets.
so, while training with the data, is it the training result how we determine the bias/threshold? how we will come to know we need to increase or decrease bias?	We don't manually decide the bias value. Training learns the bias along with the weights by minimizing the loss function. We look at validation performance:High training + high validation error means a case of likely underfitting → reduce model constraints/increase complexity. Low training + high validation error is likely overfitting → regularize more or simplify the model.
Are L1 and L2 standard terminologies for Lasso and Ridge?	Yes these are industry standard terminologies

I am totally not understanding the connection between Artificial Neural Network with Ridge and lasso, Von - Fermi equations and linear model equation, Can anyone show me the connections/bridge?	Let me try to put this in a simpler way: Linear Model: The basic engine. Fermi/Sigmoid: The steering mechanism that bends the output. Neural Network: A massive vehicle built out of hundreds of these engines. Ridge/Lasso: The brakes we install to stop the vehicle from spinning out of control (overfitting). Hope this makes sense now
who will decide threshold? and, how ensure than threshold would be the correct one?	We need to experiment, and choose the thereshold depending on our dataset
Can we use both ridge and lasso in the function?	Yes, both can be used with a linear combination ((some constant * L1) + (some other constant * L2)). However, this introduces other constants as hyperparameters. So, care should be taken while using L1 and L2 in combination.
Why is the bias equated to threshold in classification while the same bias is referred to as Offset in Regression problems? How they differ functionally?	Bias is basically always an offset. In classification, it is also called threshold for verbal clarity. The bias (offset) does act as the threshold in classification.
so hyper plane is classification or regression thing?	A hyperplane is mainly used as a decision boundary in classification
so hyper plane is classification or regression thing?	Hyperplane is used in both classification and regression but in different ways. Classification: hyperplane acts as a decision boundary separating classes. Example: SVM finds a hyperplane separating Class A and B. Regression: hyperplane can represent the best-fit relationship between features and the target especially in multiple linear regression.
when to choose classification vs regression ?	When the target is discrete for ex digit classification use classification, when the target is a continuous numerical value ex house price prediction use regression
Is there any perticular reason to plot independent variable on X axis only? Can independent variable be plotted on Y axis and dependent on X axis?	There is no strict mathematical law stopping you from swapping them, but plotting the independent variable on the X-axis is a universal convention that dates back centuries.
Can you explain what is the difference between MSE vs RMSE and when to use when?	Use MSE for optimization,RMSE is easier to interpret.

Are the models in the currently available LLM's (Basic/free versions) either overfit or underfit? because those often gives useless responses	It is difficult to say whether the models overfit or underfit without the information of the model architecture and training performance. But in general, you can expect lighter models to "underfit" as they give useless responses compared to paid heavier models. Additionally, the heavier models can also "overfit", as you may have observed in some cases where they give answers in a very different context than the context for the question. However, this "overfitting" is accompanied by memory-related errors, that will be covered in later sessions on large language models.
When should we use linear regression and when logistic regression, how to do we decide by vieweing the dataset in real examples?	Linear Regression: Could be used when the target is continuous numerical value, example house price prediction, Linea Regression could also be used as a classifier but it will only give us scores ot probablities. Logistic Regression: When we want to get the probabilities, basically a confidence score
what is the use of reducing the weights to 0. How the overfitting is getting resolved by setting the weights to 0 ?	Reducing a weight to exactly zero completely removes that input feature from the model's equation. It is the machine learning equivalent of deleting a column of useless data.
ok	
what does the formula of LR mean ?	The formula of Logistic Regression basically limits the output to the values from 0 to 1. This function can support tasks that linear regression might fail in. An example where linear regression fails but logistic works is the breast cancer diagnosis task. If you use linear regression to predict if a tumor is malignant (1) or benign (0), extreme feature values could result in meaningless predictions like 1.5 or -0.2. Logistic regression safely turns that linear score into a valid probability.
there was a discussion on ploynomial regression, where it was told as "Liner in higher order", non linear in x..could you please elaborate	Say a function is y = 2x^2. So, it is not linear when input is x. But, y linearly changes with x^2. So, y is linear in higher order of x, but non-linear in x.
If there are more than 2 class outputs then what is it called and how does it work?	When logistic regression is used for more than two categories, it is called Multinomial Logistic Regression.

what are odds	-> The "odds" measure the likelihood of an event by comparing the probability of it happening (P) to the probability of it not happening (1 - P). Mathematically, this is written as P / (1 - P). -> The "logarithm of odds", also known as the logit function, is simply the natural log applied to that ratio, written as ln(P / (1 - P)). -> In logistic regression, the model actually uses a standard linear equation (like w_0 + w_1*x) to predict this logarithm of odds. It then applies the inverse of the logit function (the sigmoid function) to transform that linear prediction into the S-curve shape, ensuring the final output remains safely inside the 0 and 1 limits.
didn't understand why need to take odds to get linear line in logistic reg?	-> The "odds" measure the likelihood of an event by comparing the probability of it happening (P) to the probability of it not happening (1 - P). Mathematically, this is written as P / (1 - P). -> The "logarithm of odds", also known as the logit function, is simply the natural log applied to that ratio, written as ln(P / (1 - P)). -> In logistic regression, the model actually uses a standard linear equation (like w_0 + w_1*x) to predict this logarithm of odds. It then applies the inverse of the logit function (the sigmoid function) to transform that linear prediction into the S-curve shape, ensuring the final output remains safely inside the 0 and 1 limits.
when is probbililty 0. Z is 0 then it is 0.5 z is infinity then p i s 1. so when is p 0 and how it is done	odds is different from the logistic regression function. Odds is p / (1-p). The "odds" is converted to logistic function using e^x / (1 - e^x)
What influences the decision whether z should be linear or polynomial?	z represents your decision boundary (the line where the model switches from predicting Category A to Category B). Use a Linear z (Straight Fence): If you plot your data and all the points for Category A are on one side, and Category B is on the other, a simple straight line can split them perfectly. Use a Polynomial z (Curved Fence): If your data is more complex, a straight line will fail. By adding squared or cubed features (x^2, x^3) to z, you allow the decision boundary to bend, loop, or curve to capture complex patterns.
Does it need to apply all these mathematical calculations in real life when some one designs the models?	No, you don't manually calculate all the math in real projects. Libraries like scikit-learn/PyTorch perform the calculations. you mainly choose the model, features, loss function and tune it.

here isnt it tends to zero from negative to 0 , it is increasing in case of a	As the x axis moves from negative value to 0, the output reduces. So the function output is decreasing.
Then where we can use those? Any scenario where we will need to look for these mathematical calculations	All these calculations happen behind the scene when we run a model using any programming tool like Python
what are weights if i am a beginner how to understand simply	You can imagine the weights as the "coefficients" in linear algebra where y = mx + c. The intercept c is the same as bias, and coefficient m is the same as weight.
After we optimize the weights, how do we fit it on the model?	We actually don't fit the weights into the model—the optimized weights are the model! When we say we have a "trained model," all we mean is that we have finally found the perfect numbers for W (weights) and b (bias).
is local minima is the just next value of global minima? can I say this?	No. A local minimum is just a point lower than its neighbors not necessarily close to or "next in line" after the global minimum. You can have many local minima scattered around and only one global minimum (the lowest overall).
why we need local minima and global minima how it will help for model to deply or train	We do not actually want local minima. They are traps! The Global Minimum is the only thing we are actively searching for. Both terms describe the "error landscape" your model navigates during training: Global Minimum (The Ultimate Goal): This is the absolute lowest possible error your model can theoretically achieve. It represents the mathematically perfect set of weights. Local Minimum (The Trap): This is a "false bottom." It is a spot where the error is lower than everything immediately surrounding it, so the model thinks it has finished the job—but there is actually a much deeper minima further away.
Why do we do double derivative? Why do we do derivation ?	1st derivative tells you which way the function is moving. It is used to find candidate points (where slope = 0). 2nd derivative tells you what kind of point it is: positive → minimum negative → maximum zero → possible saddle point In short - 1st derivative finds the point, 2nd derivative confirms what it is.

what is closed form expression? and why is it not always possible?	Closed form is a direct formula that gives you the exact answer in one step. Not always possible because complex models don't have simple algebraic solutions so we use iterative methods (like gradient descent) instead.
what value of weights do we initially take when we first run model for test, are they random and then we optimise it by running many trials?	Yes, they are generally randomly initialized and then optimized. The random distribution is normal (zero mean unit variance) or uniform distribution (-1 to 1)
Can you provide real life application of Logistics regression as compared to others regression?	Logistic regression is used when the output is categorical/binary not continuous. Real-life: spam vs not-spam, loan default (yes/no), disease diagnosis (positive/negative), customer churn prediction. Linear regression predicts a number (like price).
How to identify if the local minima is still not the global minima and will still decrease. Also how we identified that this is the global minima and nothing is lesser than that value.	There is no way to know whether the minima we obtained is actually global minima. One way to increase the chances to converge at the global minima is to train many models with different initializations, and then pick the model which performs the best on test samples, hopefully through converging at the global minima (however, there is still no guarantee of global minima convergence).
but how do a model understand the optimal global minima? i mean, how will the model know when to stop?	The model doesn't know it reached the global minimum. It stops when: Loss stops improving Gradient becomes very small Maximum iterations/epochs are reached Validation loss stops improving So, we can find a minimum but we usually cannot guarantee it is global especially in deep learning.

@Anirban Dey - Do you mean that the python library/function used to train the model, by default optimizes the weights by itself without a need of any extra code from our side	Yes, it is entirely automatic. When you use popular Python libraries like Scikit-Learn or TensorFlow/Keras, the complex optimization mathematics are completely handled for you under the hood. In most machine learning libraries, the entire training process is triggered by one simple line of code [i.e. model.fit(X, y)]. The moment you run that command, the library automatically: -> Initializes the starting weights and bias. -> Calculates the loss function (like MSE). -> Runs the gradient descent loops to push the errors to the absolute bottom. -> Freezes the optimized weights inside the model so it is ready to make predictions.
why did we add the postive number lamda here?	We add a positive number because the formula calculates Loss (Error), and our only goal is to push that total number as close to zero as possible.
With this approach there is always a chance that we can stuck in local minima .. do we do multiple iterations or random position again to again seek minimum	Correct. To avoid such cases, Starting over from random positions is exactly one of the core strategies we use.
how to know we need to increase weight or decrese weight ?	This is learnt by calculating the gradients. The gradient gives the direction (increase or decrease of parameters) for increasing the loss. Then the parameters should be moved in the opposite direction to decrease the loss function (hence the negative sign)
so gradient is same as finding slope at a point?	Yes for a single variable function, gradient = slope = the derivative at that point. For multiple variables, gradient is the vector of slopes. One partial derivative per variable pointing in the direction of steepest increase (and gradient descent moves opposite to it toward the minimum).
Gradient Descent is just for learning right?	Mostly yes. It’s an optimization algorithm used to learn the model’ s parameters (weights) by reducing the loss.

For a more complicated Slope , how does the algorithm know it is the absolute minimum ,it might have just encountered a local minima. And how do we pick the starting point	For complex cases, the answer is not straight forward. Unless you are working with a very simple model, you are entirely flying blind. Here is how it breaks down based on what you are building: Simple Models (Linear & Logistic Regression): We do know. The mathematical landscape for these models is strictly "convex"—meaning it is shaped exactly like a smooth, perfectly round soup bowl. There are zero local minima. If your algorithm finds a flat bottom, mathematics guarantees it is the one and only global minimum. Complex Models (Neural Networks): We don't know. The landscape has millions or billions of dimensions and is shaped like a highly jagged mountain range. Because the algorithm can only calculate the slope of the exact spot it is currently standing on, it can never "zoom out" to look at the entire mountain range to verify if a deeper valley exists somewhere else. In such cases, we try to evaluate the level from time to time and try to arrive at the most achievable minima
what is r	Residue/ error
Model have to increase weights to get to dL/Dw = 0 because we might started with lesser weights , might be there is scenarion where we starte with larger weights and now model havel ti decrease weights to get dL/dW = 0 , whar are those scenarios where we starte weights as low or high	We actually start with random initialization of weights and biases with normal distribution (0 mean 1 variance) or uniform distribution in range [-1, 1]. If larger or smaller weights or biases are required, they are just learnt through multiple iterations.
if theta at maxima then also derivative is zeo, in that case theta new = theta initial - (derivative = 0) which is theta new = theta initial, which is wrong	When theta is at maxima, it is something called "unstable equilibrium". It is almost impossible for the derivative to be exactly zero, meaning that slight movements can move the parameters easily from the maxima. Practically, the perfect balancing (derivative = 0) at maxima never happens.
but do we pre-define loss function graph before training model as then only will know to update learning parameter in right direction?	We pre-define the loss formula not its shape. The gradient calculation at each step tells us which way to move.We discover the landscape as we go step by step.

How would we know we have reached the global minima and not local minima while doing the optimization?	For complex cases, the answer is not straight forward. Unless you are working with a very simple model, you are entirely flying blind. Here is how it breaks down based on what you are building: Simple Models (Linear & Logistic Regression): We do know. The mathematical landscape for these models is strictly "convex"—meaning it is shaped exactly like a smooth, perfectly round soup bowl. There are zero local minima. If your algorithm finds a flat bottom, mathematics guarantees it is the one and only global minimum. Complex Models (Neural Networks): We don't know. The landscape has millions or billions of dimensions and is shaped like a highly jagged mountain range. Because the algorithm can only calculate the slope of the exact spot it is currently standing on, it can never "zoom out" to look at the entire mountain range to verify if a deeper valley exists somewhere else. In such cases, we try to evaluate the level from time to time and try to arrive at the most achievable minima
How the Lasso and Ridge become a part of this gradent descent process and weight decison process?	Lasso and ridge regularization losses also support gradients and backpropagation, so they work exactly similar to any other loss. The loss landscape is already shown in previous slides (the V shape for L1 and U shape for L2).
can you give an example of optimization	So let’s say you're a delivery company trying to minimize fuel cost. Function to optimize: total distance/fuel used Variables: the route/order of stops Optimization = trying different routes and adjusting until you find the shortest/cheapest one
what is lamda	It is a number you set before training begins that controls exactly how strict the complexity police are going to be.

what value does lamda substitue for here?	It is a hyperparameter—a blank variable that you (the programmer) get to choose and type into the code before training even begins. New Total Loss = Standard Error (like MSE) + lambda *(Size of the Weights) Because it is a multiplier, the numeric value you substitute for lambda dictates exactly how heavily that penalty affects the final score.
gredient descent sir	Take an example of the U shaped loss function of L2 regularization, and V shaped loss function of L1 regularization. Basically, for a parameter, when it is a large value, it gets a loss and moves toward the 0 value. This is done with gradient descent. Imagine the loss function as a hill. Then gradient descent is the gravity that makes the parameter move to a lower height of hill. How it is done is with mathematics that the professor explained.
what r represents here?	r is the learning rate. It dictates the value of movement. For example, larger r value makes parameters to change faster, whereas lower r value makes parameters to change slowly.
Can you give examples of the all math terminogies used today. For example optimization : delivery company trying to minimize fuel cost. You can provide answers later in resources	sure
Here in our examples, one model = one equation. But for LLMs, one model is not equal to one equation, as we can ask LLM to do a lot many things starting from answering general queries to analyze the uploaded data. So, what does a model mean practically for LLMs?	LLMs are advanced models and will be discussed in details in the later sessions. However, just to give an idena, LLMs use concepts from this session (gradient descent and loss function) along with additional concepts (short term memory, long term memory, self-attention, multiple models where each one is expert in a specific task etc).
for logistic regression, is ylog(f(x)) + (1-y)log(1-f(x)) the loss function	It's mostly called Log Loss. In machine learning, algorithms like Gradient Descent are strictly built to walk downhill to minimize error.
What is Practical Real-world example of Liner and Logistic Regression??	Linear Regression → house price prediction. Logistic Regression →spam or not-spam email detection.
if we have non linear function we can define XOR boundary right?	Correct, you must use a non-linear function to solve the XOR problem.

Sir, you explained about parametric, what about non-parametric models ? When do we use one vs another ?	The choice comes down to how much you trust the shape of your data and how much data you actually have. Parametric Models (The "Off-the-Rack" Suit): Use it when you have a smaller dataset, you need blazing-fast training times, or you already know the data follows a simple, predictable trend. Non-Parametric Models (The "Custom-Tailored" Suit): Use it when you have plenty of data, the data has highly complex or hidden patterns (like the XOR problem), and you care more about pure accuracy than having a neat, readable math formula.
Please share the link to notebook	Colab notebook link: https://colab.research.google.com/drive/1-wSa5t-1C2Ibvz_gefSgrtP1jIRH5QKP?usp=sharing Dataset link: https://drive.google.com/drive/folders/13eGeeE7mUgI0d8hK5vRnWxNFDb_6imwn?usp=sharing
is this SKlearn standard Lib or Custom Created ?	it is a standard built-in library not custom-created
Is median best metrics for Outlier treatment?? or there are any other best standards??	Median is one good robust option for replacing outliers but IQR and Z-score are more standard for detecting them first. Best method depends on data distribution and use case.
Sir, shouldn't we dothe outelier masking on all data before train test split just in case test might contain more outliers than train set ?	No. Test set is used to mimic real life data. So after training happens, we do not have information about the outliers in real-world samples. So we cannot train according to them. Similarly, ONLY the training data should be used to deduce any preprocessing to be done. The test cases should be preprocessed just the same as the training cases are preprocessed, but should not be used to decide the preprocessing.
what are some of the best OL methods then?	For quick use — prefer IQR or Z-score. For high-dimensional data — use Isolation Forest or DBSCAN.
Can you giude on some best practices to prevent data leakage other during training ?	A good way to prevent data leakage is to separate the training and testing samples right from the beginning. If only the training samples are used to decide any preprocessing, there is no data leakage.
How is threshold is used as 0.5??	The two classes are defined as 0 and 1, so it is expected that if the separation is done at 0.5, then the output will be properly separated.
Can you add the comments next to StandardScalar() to show why it is being used again? It will be helpful to know the reasons for each step we are taking.	Good suggestion. Just to clarify, StandardScalar() changes the numerical data to a mean of 0 and standard deviation of 1, for stability in training.

Can you please repeat what is Degree 1, 2, 3 here?	It Refers to polynomial degree i.e. how many powers of features you add: Degree 1 = linear (straight line) Degree 2 = adds squared terms (curve) Degree 3 = adds cubed terms (more flexible curve) Higher degree = more complexity = higher overfitting risk
If we are to extend this problem to different stages of cancer like stage 1, 2 ,.. how shall our threshold be defined ?	There will be multiple thresholds in that case. For example, 0.5, 1.5, 2.5 and so on, to separate stage 1 from the rest, then to separate stage 2 from the remaining rest, and so on.
How the alpha value was decided. is it empirical	Yes. Whether you are using Ridge (L2) or Lasso (L1), the alpha is empirical.
Can you please give some real-world use cases of Linear, Polynomial and Logistic regression?	-> Linear Regression: Used when predicting a continuous output that scales proportionally with the inputs. A practical use case is predicting the expected time-of-flight or signal delay for an acoustic wave in ultrasound beamforming based on the linear distance from the transducer element. -> Polynomial Regression: Used when the continuous output follows a curve rather than a straight line. A common example is modeling the non-linear acoustic attenuation of high-frequency waves as they penetrate deeper into biological tissue, or computing the curved parabolic trajectory of a jumping character in a custom 3D game physics engine. -> Logistic Regression: Used strictly for binary classification where the output must be squashed into a 0 or 1 probability limit. Beyond the breast cancer tumor diagnosis we already looked at, another strong use case is analyzing a reconstructed low-dose CT scan to classify whether a specific anatomical region contains a malignant anomaly (1) or is completely healthy (0).

what is the use of regularization term? we could have achieved minimum loss via the MSE right? why was this extra term needed?	if our only goal was to get a perfect score on the data we already have, MSE is all we need. But our goal isn't to get a perfect score on the data we already have. Our goal is to make accurate predictions on data we haven't seen yet. If you only use MSE, the algorithm becomes completely obsessed with hitting every single data point perfectly, including the random noise, typos, and outliers. It will twist its equation into absurd, complex shapes just to force the error down to exactly 0. Example: Imagine a student studying for a history test using a practice exam. MSE Only: The student literally memorizes the exact sentences on the practice exam, including a typo on question #4. They get a 100% on the practice exam (Minimum MSE). But when they take the real exam with slightly different questions, they fail completely because they didn't learn the actual concepts. MSE + Regularization: In this, the student learns the overall concept and answers questions that was previously unseen
While transforming Train and test, why didn't we fit.transform test data set?	We fit only on train not test to avoid data leakage. Test data should stay "unseen" so we reuse train's learned parameters and just transform test with them — fit_transform() on train transform() only on test.
wrt what sir is explaining, our target should be to have training score < test ? Wouldn’t that be underfitting ?	No, in case of underfitting the model performs bad on both the training and testing sets. When training performance is much better than testing performance, it is an indication that the model has overfitted. In the explanation, both things are needed: 1. the performance is good on both train and test sets (no underfitting), 2. the testing performance is (even if slightly) better than training performance (ensuring no overfitting).
When will the assumptions to build linear regressions or any other models will be thought?	Assumptions are build during EDA phase. It helps you pick the right model not after building it.
Can you please help understand what are the scenarios when we go for Ridge and Lasso? Why is it so important? Is it mandatory to do in every linear model?	L1 is used when the number of parameters need to be reduced, for example in edge devices, as it reduces many of the weights to 0. L2 works better in general when no reduction in parameters is needed. While it is not mandatory to do L1 or L2 regularization in every linear model, the presence of L1 or L2 regularization does make the training more stable.

Instead of Lasso why cant we use the corr matrix in the pre processing and remove correlated features. What could be preferred option	You actually can use a correlation matrix to remove features, and you usually should! But they are not replacements for each other because they evaluate your data in two completely different ways. The preferred option in the real world is to use both in a two-step process.
On what basis does the Lasso regression makes some weights zero?	Lasso panelizes even the small parameters, forcing most of parameters to 0. Ridge only panelizes the large parameters, so the small parameters are not forced to zero.
Does L1 and L2 means only for regularisation in a LOSS equation?	Yes. L1 and L2 are added in loss function only.
at the end ,how do we decide recall is not required?	We decide to ignore Recall when the business cost of a False Alarm (False Positive) is completely unacceptable, but the cost of Missing Something (False Negative) is barely noticeable.
So, everytime we have to follow these process ? Linear, L1, L2 and LR?	Not necessarily, this was just for demonstration purpose, but you definitely need to experiment and find what works best for your situation
Can you explain FP, FN, TP, TN?	To put it in a simpler way: TP: You guessed Yes, and you were right. TN: You guessed No, and you were right. FP: You guessed Yes, but you were wrong. FN: You guessed No, but you were wrong.
Looking at current dataset example does it mean that we don't need to understand the data relationship ourself ?	Yes you are right. The code shows good ways to automatically deduce the relationship of data and preprocess accordingly. However, care should still be taken to ensure preprocessing steps are not done inaccurately (which is a common mistake in training AI models).
Practical example of choosing L1 over L2	When dataset has multicollinearity or in simple language features are not highly correlated prefer L1 else L2
I'd like to understand w/ an example preferably where Y = AX1 + BX^n . How does the inference play out? How can we or use teschniques to get which variable influenced Y. Share some pointers	Imagine predicting house prices: Y = A(X1) + B(X2)^2 Where: X_1 is Number of Bedrooms (linear). X_2 is Distance to City (squared, because commuting pain scales exponentially).

Does the model selection depends on how the feature is dependent on labelled output? Like if some features are linearly dependent on output variable while some are non-lineraly dependent on output variable. what model to be used?	You are correct, model selection does depend on the data. When using simple models, model selection is necessary. However, later sessions will discuss more complex "universal function approximators", that can handle both linear or non-linear data. However, they take more time and computations.
From the earlier class, I came to a conclusion that regression is more of to get the the numerical answer, but in this class linear regression mostly focused on classification and the logistic regression did the same. I got a little confused between regression and classification. Can you please share some knowledge insights to clarify this.	Regression = predicts a number. Classification = predicts a category. Linear Regression → regression (predicts numbers) matches its name Logistic Regression → despite the name, it's classification (predicts categories via probability + threshold) Key point: "Regression" in the name doesn't always mean regression task
forecast the next month sales?	Right. An example where we are trying to predict numerical value, and it calls for Linear regression
forecast the next month sales?	But normally, it's not so simple in the real world. In most cases it'll will be non-linear and needs to be handled accordingly
is there any loss function which is best for all models	While there is no "best" loss function, mostly two loss functions work in most cases: Mean Squared Error loss for continuous numerical values and Cross Entropy loss for classification tasks.
but during preprocessing of data we are already dropping highly correlated data. Then we will not have that much of multicolinearity in my dataset. then how L1 will be choosen?	Removing correlated features handles multicollinearity. L1/Lasso is chosen separately for automatic feature selection not just multicollinearity.
You have uploaded the csv and choose the training 80% data and 20% data for test. so if i upload and query similar data (with same columns) , do we need to again train the data ?	No, you can just infer. After the training has happened, it is ready for the real world (given that training and testing accuracy is good). Just ensure that the data is exactly like when training and testing (number of inputs, data type of inputs, correct features used in correct index positions).
As we can have multiple local minima, and a single global minima, how would gradient descent	The gradient descent happens to reach the closest minima in general. It is like moving in a hilly area blindly. Wherever the gravity takes you, you reach that corresponding valley, not necessarily to the lowest elevation valley.
As we can have multiple local minima, and a single global minima, how would gradient descent lead to selecting global minima	There is no way to guarantee convergence to a global minima. One way to counter this is to use many models with different initializations, and pick the best performing model, that is expected to be at the lowest minima.

What is the difference between Linear regression, OLS regression and non-linear regression models	Let me try to explain with examples: Linear Regression (The Concept): You assume a barista's base pay goes up at a constant rate based on hours worked. Every extra hour equals exactly $15. It is a strict, straight-line rule. Use linear regression OLS (The Tool): You look at the barista's actual paychecks, which are random because of random daily tips. OLS is the specific mathematical calculator used to draw the "fairest" straight line right down the middle of those messy data points to figure out that $15/hr average. Non-Linear Regression (The Curve): Now look at the barista's caffeine intake. Cup 1 wakes them up massively. Cup 2 helps a little. Cup 6 makes them jittery and crash. The effect is not a straight line—it curves and drops. If you used Linear Regression here, the math would wildly predict that drinking 100 cups of coffee grants infinite energy. Non-Linear regression allows the math to bend to capture that real-world crash.
the learning rate would play a major role in converging to minima right?	Yes , learning rate controls step size in gradient descent. Too high indicates overshoot/diverge while too low indicated slow convergence. Getting it right is key.
Is there some real world case where we would use both Ridge and lasso together?	In general, no. If you have storage and computation constraints, use L1, otherwise use L2. However, it is practically possible to use a combination of the two.
On what basis does the L1 norm considers a feature isnt useful and assigns the weight to 0? how does this underlying feature works?	L1 doesn’t explicitly decide that a feature is useless. It penalizes large coefficients and during optimization, some coefficients are pushed exactly to 0. A coefficient becomes 0 when its contribution to reducing the prediction error is smaller than the L1 penalty. A zero coefficient effectively means that feature is excluded from the model.
but the code shown now have histogram code, training code. for real data , where i want to directly predict (already model trained) the output then whether those code nee to be optimized or do we need to remove some part of code?	When inferring on new real-world data, a lot of the code is not needed. The entire training part (except the testing part) needs to be removed. The visualizations can also be removed if not needed.

When we say features, We basically are indicating the columns coming as apart of the dataframe? does handling a categorical field in terms of number does increases the feature count ?	Yes. A "feature" is simply the academic word for a column in your dataset (excluding the final answer you are trying to predict, which is called the "target" or "label"). As for handling categorical fields, yes, it usually increases your feature count
Also what is the way to understand if a model is overfitting or underfitting by looking at confusion matrix, feature importance, prediction result (classification model)?	If both training and testing performance low, then underfitting. If training performance is much better than testing performance, then overfitting. If testing performance is better or comparable to training performance, and both performances are also good, then no underfitting and overfitting in general.
For the coding part for any problem statement, what should be our generalized approach to solve it? Example : I have a problem statement of Cancer detection, it can be anything, What should be my solution approach to identify whether I have to use linear regression or logistic regression and after that what are the steps I need to keep in my mind to proceed further. I can go through the notebooks for thee 2 sessions and see what steps we followed in 2 cases but in general how do we need to picture it while solving any problem statement.	So the first step is problem statement formulation, if its a classification problem or a regression problem. First step is exploratory data analysis, then we prerocess the data based on EDA, then if its a classification we use classification models (as of now we only know of Logistic Regression), then if regression we fit Linear Regression. Now we need to look if the model is underfitting or overfitting. And regularuze accordingly
got it, we have considered only one case (cancer type), if it's more than one case, then how to differentiate when the data or csv is loaded? can we train the same variable or model? can u share some inference or link to understand	There can be two different cases here, one is for example a cancer type of 1, 2, 3, and so on (this is called mutually exclusive). The other can be cancer vs not cancer, and additional separate classes like ultrasound vs CT (this is called non-mutually exclusive). These cases will be covered in later sessions, but in short, we use Categorical cross entropy loss for the first case, and Binary cross entropy loss for each label ("cancer-ness", modality) for the second case.
Are there multilevel classifiers	There can be two different cases here, one is for example a cancer type of 1, 2, 3, and so on (this is called mutually exclusive). The other can be cancer vs not cancer, and additional separate classes like ultrasound vs CT (this is called non-mutually exclusive). These cases will be covered in later sessions, but in short, we use Categorical cross entropy loss for the first case, and Binary cross entropy loss for each label ("cancer-ness", modality) for the second case.
Thanks Raj.. any reference or link for AI/ML coding...based on my queries	The tutorial coding notebook actually serves as a great resource for coding reference. You can also follow youtube tutorials from "Aladdin Persson" for many practical "from scratch" implementations if needed.